Promoting Long-tail Items through Debiased Product Bundling

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Product bundling organizes thematically related items into combined bundles to facilitate shipment and promote products. Sellers typically bundle fresh or overstocked items with popular ones to increase the exposure of these items. This specific task can be formulated as a long-tail product bundling scenario, where item popularity is defined via user-item interactions. Conventional methods suffer from inherent popularity bias in pre-extracted user feedback features and insufficient utilization of popularity-independent knowledge, which forces them to prioritize popular items and struggle with long-tail bundling. We propose the **D**istilled Modal**i**ty-Oriented Knowl**e**dge **T**ransfer (DieT) framework to counter this bias by maximizing the mining of popularity-free features while preserving real-world bundling intent. Specifically, DieT employs two modules: (1) Popularity-free Collaborative Distribution Modeling (PCD), which captures intrinsic bundle-item relationships independent of popularity signals, and (2) Unbiased Bundle-aware Knowledge Transferring (UBT), which utilizes knowledge distillation to amplify these features and suppress user feedback-induced bias. Extensive experiments on two real-world datasets demonstrate DieT's superiority over state-of-the-art methods in long-tail bundling, achieving statistically significant improvements through the effective integration of popularity-agnostic knowledge. This work addresses critical limitations in automated bundling systems by balancing inventory-driven requirements with unbiased feature representations. The code has been released at https://github.com/Shuo-Xu-0319/DieT.

CCS Concepts: • **Information systems → Recommender systems**.

Additional Key Words and Phrases: Bundle Construction, Multimodal Modeling, Knowledge Distillation

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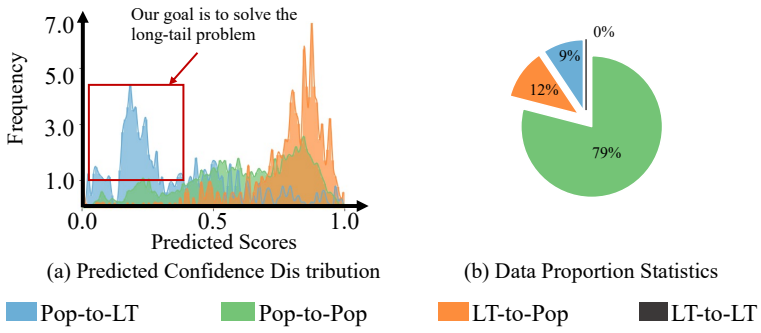

Fig. 1. (a) The predicted confidence distribution of SOTA method CLHE on four bundling settings (distinguished by colors) and (b) the data proportion statistics. "Pop" refers to popular items, while "LT" stands for long-tail items. Notably, the Pop-to-LT scenario is the most challenging with its remarkable data proportion.

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1 Introduction

Product bundling aims to curate a set of items closely related to the same theme for sale as one combined bundle [1, 3], which has been widely applied in e-commerce, online entertainment, and *etc.* [37, 38]. Its primary purpose in these real-world online applications is to increase the exposure of overstocked products, which are often characterized as long-tail items *w.r.t.* user-item interactions. Specifically, these long-tail (abbreviated as LT) items are purposefully bundled with certain popular (abbreviated as Pop) items [1] which are commonly regarded the most crucial in online serving, which are defined as Pop-to-LT product bundling scenarios [2].

Despite the significance of bundles in various online applications, automatic product bundling has received disproportionately limited attention in both academia and industry. CLHE [40] serves as the pioneering work to explore this important but under-explored frontier via effectively utilizing bundle-item affiliation, user-item feedback, and multimodal content knowledge. Experimental results indicate that CLHE significantly outperforms previous bundle-related studies, such as Bi-LSTM [11], Uhbr [63], and Bruce [57]. By introducing multi-level self-attention mechanisms to merge multimodal features and leveraging contrastive learning to mitigate noise, CLHE achieves superior performance in the product bundling task. However, mechanically integrating these three multi-view information in CLHE might yield results that deviate from the original intent of bundle construction. Moreover, the introduction of user-item feedback information in overall product bundling performance has led to unequal treatment of long-tail items and shows limited performance in the specific Pop-to-LT scenario, as shown in Figure 1 (a).

Scrutinizing this problem, we conduct a systematic and empirical study across the product bundling scenarios and evaluate the impact of various features from diverse views. Specifically, we surgically break down the bundling setting into four sub-settings, *i.e.,* LT-to-Pop, Pop-to-LT, Pop-to-Pop, and LT-to-LT, based on the popularity distribution of the items in both query and output

[1]The popularity discussed here is solely related to the interaction number between users and items,
[2]Pop-to-LT: Given some popular items as query, we aim to find some long-tail items to complete the bundle (*e.g.,* attract customers with a trendy dress and pair it with a less popular yet matching handbag).

partial bundle. Then we apply CLHE [40] on these sub-settings to obtain its predicted confidence scores on the ground truth item with the given partial bundle. By aggregating the data proportion statistics for each scenario, we visualize the comparative results in Figure 1. Investigating these sub-settings, we found that CLHE achieves lower performance in the Pop-to-LT setting compared to other settings since it has the lowest confidence in the predicted confidence distribution of Figure 1 (a). At the same time, retaining a non-negligible portion of cases as shown in the data distribution of Figure 1 (b) [3].

Zooming into the Pop-to-LT setting, based on our observation, a natural question arises: *what are the key factors that cause the deteriorated performance?* To answer this, we identify two potential reasons by intuitive and empirical analysis as follows. First, ***item-level user feedback features are not well aligned with the specific objectives of the Pop-to-LT bundling scenario***. These features are pre-extracted from collaborative filtering on the user-item view, which may force the model to prioritize more popular items [12, 40]. The indiscriminate incorporation of these features increases the ranking of popular items, hence leading to disastrous low-ranking for low-popularity items. Further surgical analysis (see Section 3.3) empirically demonstrates that removing the user feedback features will not hinder the performance; in fact, it can even yield performance gains specifically within the Pop-to-LT setting. Second, ***the popularity-free features,* i.e., *bundle-item affiliation and item multimodal features, are not maximally utilized***. Interestingly, we find that removing bundle-item affiliation or item multimodal features results in a significant performance drop in the Pop-to-LT setting, justifying our hypothesis. In summary, the crux to tackling the Pop-to-LT problem lies in maximally mining the popularity-free features and effectively integrating them into the bundling process.

To address the aforementioned issue, we propose a novel **D**istilled Modal**i**ty-Oriented Knowl**e**dge **T**ransfer framework (DieT). By employing a teacher-student distillation paradigm, DieT is designed to maximally leverage popularity-free signals (specifically, bundle-item affiliations and multimodal content), thereby serving as an effective remedy for the marginalization of long-tail items. DieT employs knowledge distillation to mitigate item popularity bias within the product bundling task, thereby fulfilling the core objective of leveraging popular items to drive the inclusion of long-tail items in the bundling process. Our model consists of two main modules. A pre-trained Popularity-free Collaborative Distribution Modeling module (PCD) serves as the teacher model, learning the collaborative knowledge between bundles and items. At the same time, the Unbiased Bundle-aware Knowledge Transferring module (UBT) serves as the student model, training with the bundling task using multimodal knowledge, and optimizing content features guided by the collaborative knowledge from the teacher model. Specifically, first, based on the experimental analysis, we highlight the essential role of defined popularity-free knowledge in the Pop-to-LT scenario. DieT designs PCD to capture the bundle-item hierarchical collaborative distribution through an intra-view self-attention mechanism. Second, to incorporate popularity-independent information into the Pop-to-LT scenario, DieT further proposes UBT to enable the directional knowledge transfer from the bundle-item view to the multimodal view through the knowledge distillation paradigm. This mechanism strategically mitigates the inherent popularity bias rooted in user feedback, thereby rectifying the under-representation of long-tail items. Meanwhile, this knowledge distillation framework maximizes the model's performance in the overall scenario.

To evaluate our framework, we conduct extensive experiments on two real-world datasets. We compare several product bundling approaches and evaluate the results of adopting debiasing techniques from traditional recommendation systems to the product bundling task. The results show that our method significantly improves performance in the Pop-to-LT product bundling scenario.

[3]The LT-to-LT setting has few data while the LT-to-Pop and Pop-to-Pop settings already perform pretty well.

In-depth analysis and case studies further demonstrate the high generalization and robustness of our method in diverse settings. The contributions of this work can be summarized as follows:

- We start with the original purpose of product bundling and divide it into four scenarios. Then, we analyze the results of the current method for each scenario. To the best of our knowledge, we are the first to identify the scenario of Pop-to-LT product bundling with extensive intuitive and empirical study for the sake of long-tail item promotion.
- We conduct a series of empirical studies and surgical analysis in the Pop-to-LT scenario and examine the critical reasons for inferior Pop-to-LT performance as being the inadequate utilization of popularity-free features, based on which we propose a novel model-agnostic knowledge distillation-based framework DieT.
- Our method introduces a teacher-student model distillation framework that improves the utilization of popularity-free features through collaboration between the two models, thus addressing the popularity bias problem in product bundling. DieT achieves significant performance gain over the SOTA methods in the Pop-to-LT bundling scenario, and shows comparable performance in the overall evaluation. Further analysis demonstrates its robustness and generalization capabilities.

2 Related Work

We review the relevant literature for this work, including a summary and categorization of bundle-aware tasks, as well as a brief overview of knowledge distillation and popularity debiasing in recommendation.

2.1 Bundle Recommendation and Generation.

In recent years, bundle recommendation [41, 44] systems have attracted widespread attention from both academic and industrial communities. The goal of bundle recommendation is to provide users with a group of related items instead of individual ones, meeting diverse user needs in a one-stop manner while boosting sales [2]. The bundling task spans various application domains, including fashion outfit [1, 6], movie list [14, 61], music playlists [13, 48] and games package [46]. Inspired by the significance of bundles in real-world applications, several recent studies [27, 49, 51] have reviewed and analyzed bundle-aware tasks from various perspectives. We summarize bundle-aware tasks and classify them into two types according to bundling strategies: discriminative bundle recommendation (bundle recommendation) and generative bundle recommendation (personalized bundle generation and product bundling).

Bundle recommendation focuses on recommending predefined bundles, such as meticulously crafted e-commerce product combinations or curated playlists. These models determine whether a bundle is relevant to and appealing for a user based on their historical interactions and preferences. BRP [70] first tackles this issue by constructing expected reward vectors and cross-dependency matrices. BGCN [3] decouples user preferences into both item views and bundle views to effectively capture dual user preferences. CrossCBR [38] further emphasizes the importance of collaborative association in BGCN [3] using the cross-view contrastive learning paradigm to enhance the self-discrimination of representations. MultiCBR [37] adopts an "early fusion, late contrast" strategy in bundle-item affiliations, enhancing user preference modeling while minimizing extra costs.

However, predefined bundles may not always meet the needs of users or the bundle service providers. Unlike discriminative bundle recommendation, generative bundle models aim to create new bundles for users (personalized bundle generation) or construct bundles based on partial seed items (product bundling). BYOB [7] formulates the bundle composition problem as a personalized combinatorial optimization task over a candidate set. It designs a specialized reward function within a deep Reinforcement Learning (RL) framework to address the interaction sparsity issue between users and bundles. Conna [56] focuses on generating personalized creative bundles for given users.

It introduces a contrastive non-autoregressive decoder for bundle creation, optimizing both the quality and efficiency of generation. BundleMCR [14] defines personalized bundle generation as an innovative multi-round conversational procedure and leverages the RL paradigm to model it as a Markov decision process problem with multiple agents. AICL [50] deduces the underlying intentions rooted in users' behaviors within a session, employing a large language model to execute the personalized bundle generation process.

With the explosive growth of the item corpus, manually constructing large-scale bundles is quite formidable. Thus, an automatic product bundling strategy is indispensable and has proven to be valid. CLHE [40] is the pioneering product bundling strategy that constructs the partial bundle with multimodal features, user feedback at the item level, and existing bundles, achieving the performance of SOTA construction. LARP [48] integrates multimodal and relational signals into feature representations and tackles the playlist construction task through a three-layer contrastive learning framework. CIRP [39] leverages cross-item contrastive loss and individual item's image-text contrastive loss to embed modeling capabilities into a multimodal encoder, producing enhanced item representations for downstream product bundling applications. RaMen [43] models the bundle construction process via multi-strategy and multi-modal learning, jointly capturing intrinsic semantic properties and extrinsic collaborative relations of items to infer the latent intent behind bundle formation. Driven by the powerful text-handling abilities and vast external knowledge of LLM, BundleMLLM [33] introduces a novel multimodal large language model framework that integrates multimodal features of items and formulates the bundle construction task as a multiple-choice question with candidates as options, which could harness the extensive external knowledge of LLMs to facilitate the bundling process.

Nevertheless, current efforts primarily focus on addressing the cold start problem and enhancing the overall quality of bundling construction, the pre-extracted item-level user feedback features emphasized in CLHE may intensify the popularity bias in the user-item view, hindering alignment with the bundling objective of this specific Pop-to-LT scenario.

2.2 Knowledge Distillation in Recommendation.

Deep neural networks have been proven effective in both academia and industry, particularly in fields such as computer vision and natural language processing. However, these networks often rely on a large number of model parameters to process data, resulting in high memory requirements and low runtime efficiency in practical applications. As a classic model compression approach, the core of knowledge distillation (KD) is that a large teacher model injects hidden knowledge extracted from the data into a simpler student model, thereby enhancing the student model's performance in a specific aspect [16].

Inspired by its success in computer vision [28], natural language processing [32], and information retrieval [30], KD has been widely applied in the field of recommender systems to reduce model complexity or enhance recommendation efficiency [21, 23, 52]. In [52], the concept of knowledge distillation is first applied to large-scale ranking tasks. It achieves enhanced online inference efficiency by using a high-parameter teacher model to guide the training of a compact student model. ALDI [17] designs three distilled aligning objectives to transfer behavioral information from warm items to cold ones for cold-start problems. SGFD [31] proposes response-based and feature-based distillation losses to facilitate the rich modality features transfer in multimodal recommendation. CrossDistil [62] proposes a quadruple-loss-based knowledge distillation framework for multitask learning in recommendation systems. It enables cross-task knowledge extraction by transferring knowledge from an auxiliary task (teacher) to the original task (student). CausalD [65] leverages causal theory to investigate the origins of performance heterogeneity in recommendation systems.

By proposing a multi-teacher causal distillation framework, it distills the causal effects from heterogeneous teachers into a unified recommendation model, enhancing recommendation effectiveness and addressing performance heterogeneity without the need for unbiased data.

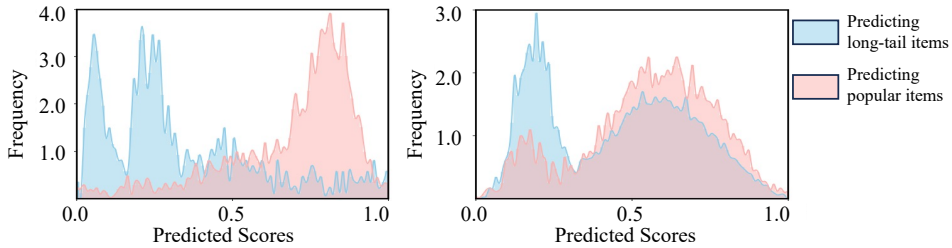
While knowledge distillation is widely utilized in recommendation systems, its application within the bundle domain remains insufficiently explored. DGMAE [47] is the most relevant study to our work, which extracts the first- and higher-order associations from the user-bundle view to identify users' potential bundling preferences in bundle recommendation tasks. To the best of our knowledge, DieT is the first KD framework for product bundling tasks.

2.3 Popularity Debiasing in Recommendation.

Current recommendation system research mainly focuses on providing users with more accurate and personalized recommendation services. However, user behavioral data is collected from observational rather than ideal experimental settings, which inevitably introduces multiple types of bias into recommendation models, including position bias [9, 18], selection bias [25, 54, 55], and popularity bias [5, 66, 69]. This feedback loop further amplifies popularity imbalance, resulting in the well-known "Matthew effect," where already popular items continue to gain exposure while long-tail items receive even less attention [53]. For traditional recommendation tasks, the previous efforts typically utilize additional regularization [34, 71], causal inference [10, 66], and auxiliary knowledge [35] to address this issue. PPAC [45] proposes a user-perceived item popularity model for different users, designing a model-agnostic, personalized counterfactual framework. By integrating personal and global popularity, and using counterfactual methods, it effectively reduces global popularity bias. MDTF [64] designs two modules, the General Ranking Model and the Popularity Debias Module. By estimating the impact of item competitiveness and popularity on click-through rates, it reduces popularity bias while improving model performance. APDA [69] systematically analyzes the causes of popularity bias in GNN-based collaborative filtering. It proposes a new graph aggregator that uses adaptive weights and strengthens residual links, reducing popularity bias while maintaining ranking performance.

Previous studies have observed that user–bundle interactions typically exhibit a highly skewed distribution, where popular bundles dominate exposure while less popular or newly introduced bundles are easily neglected [49]. To mitigate this issue, a growing body of work has explored popularity-aware and long-tail–oriented modeling strategies in bundle recommendation. PopCon [19] proposes to reduce popularity bias by incorporating popularity-based negative sampling during training and applying a configuration-aware re-ranking strategy to improve diversity at inference time. In the same year, CoHeat [20] introduces a multi-view framework that jointly models user–bundle and user–item interactions. To address the imbalance of bundle popularity, CoHeat emphasizes the user–item view for unpopular bundles, while curriculum learning and contrastive learning are leveraged to stabilize representation learning across views. More recently, several works further deepen the understanding of popularity bias and long-tail issues in bundle recommendation. A reproducibility study [42] systematically investigates product-side fairness, demonstrating how popularity bias in historical interactions propagates through bundle recommendation models and leads to exposure unfairness at both bundle and item levels. CALBRec [36] alleviates the long-tail bundle recommendation problem by mining shared compositional patterns across bundles, adaptively strengthening tail bundle representations while suppressing noise from sparse interactions. GAMNBRec [24] addresses the imbalance between popular and long-tail items through adaptive graph normalization, residual-enhanced dynamic feature fusion, and a weighted contrastive BPR objective, achieving improvements in both recommendation accuracy and diversity.

Notably, these works address popularity bias in user-bundle interactions by treating bundles as items, similar to traditional cold-start recommendation tasks. However, the primary purpose



(a) POG (b) Spotify

Fig. 2. The score distribution for predicting long-tail and popular items of CLHE on two datasets. The horizontal axis shows the item scores, and the vertical axis represents their occurrence frequency. In both datasets, when the target items are long-tail items, the results are suboptimal.

of bundling is to combine several related items, using popular products to boost the exposure of other items in the bundle. Thus, the issue of item popularity within a bundle becomes a critical research focus, especially in bundle generation tasks. Our DieT emphasizes the popularity of each item, achieving the promotion of long-tail items through the in-depth extraction of popularity-free knowledge and distilled knowledge transfer.

3 Preliminary and Empirical Study

We first introduce the product bundling task, with the specific definition of the Pop-to-LT scenario. Then, we conduct empirical studies to indicate the issues with existing methods and highlight the motivation for this work, including performance analysis of each scenario and surgical analysis of multiple modalities.

3.1 Problem Formulation

We aim to maximally capture the popularity-free knowledge and effectively transfer it into the bundling process to enhance the Pop-to-LT product bundling scenario. To achieve this, we first define the set of items as $\mathcal{I} = \{i_1, i_2, \cdots, i_n\}$ with the corresponding textual features $\mathcal{T} = \{\boldsymbol{t}_1, \boldsymbol{t}_2, \cdots, \boldsymbol{t}_n\}$ and media feature $\mathcal{M} = \{\boldsymbol{m}_1, \boldsymbol{m}_2, \cdots, \boldsymbol{m}_n\}$, where n denotes the quantity of items. The pre-extracted textual features are obtained from the relevant descriptions, titles, and attributes, and the media features are extracted from the visual or audio information. Additionally, we collect the interaction data between users and items to form the user-item interaction matrix $D = \{d_{ui} | u \in \mathcal{U}, i \in \mathcal{I}\}$, where $\mathcal{U} = \{u_1, u_2, ...u_m\}$ denotes the set of users and m is the number of users. Here, $d_{ui} = 1$ indicates the presence of an interaction between user u and item i, while $d_{ui} = 0$ denotes absence. For the product bundling task, we define the bundle $b = \{i_1, i_2, \cdots, i_{\widetilde{n}}\}$ as a set of several relevant items, where $\widetilde{n} = |b|$ denotes the number of items in bundle b. Given a partial bundle $b_x \in b$, where b_x is a subset of b and adheres $|b_x| < |b|$, the goal of this task is to predict the missing item $i \in b \backslash b_x$ to complete the original bundle b. We define the specific setting where all items in b_x are high-popularity, and items in $b \backslash b_x$ are long-tail as the Pop-to-LT product bundling scenario.

3.2 Performance Analysis of Each Scenario

Due to differences in category, price, and brand, items in the corpus show diverse levels of popularity, with the presence of the "rich get richer" Matthew effect further intensifying this imbalance. However, existing product bundling methods overlook this issue, thereby unfairly treating items with varying popularity. It is intuitive that ***they significantly suffer from the long-tail problem,***

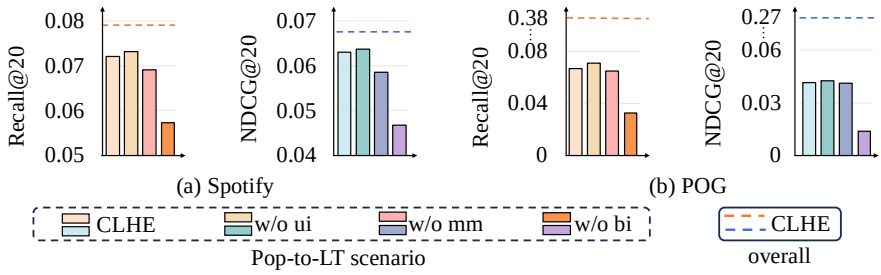


Fig. 3. Surgical results of multiple modalities in CLHE on Pop-to-LT scenario. Different colors represent the removal of specific features, and the dashed line indicates the performance of CLHE in the overall scenario. Removing bundle-item affiliation or multimodal features results in a significant drop.

resulting in poor accuracy for items with limited interactions. To verify this opinion, we visualize the predicted probability distribution of the SOTA product bundling method [40] when predicting long-tail items and popular items on POG and Spotify. Figure 2 indicates that the accuracy of CLHE is significantly lower when predicting long-tail items compared to popular ones. Notably, CLHE shows the most significant performance differences between different popularity settings in the fashion e-commerce dataset POG compared to the music-sharing dataset Spotify, which focuses more on item popularity. This phenomenon aligns perfectly with the intuitive assumptions of the Pop-to-LT product bundling scenario. In practical applications, this is a challenging and worthwhile issue to address.

3.3 Surgical Analysis of Multiple Modalities

As mentioned in Section 3.2, existing bundling strategies exhibit diverse accuracy when predicting items with different levels of popularity. In that case, it may encounter a performance collapse in the more challenging Pop-to-LT product bundling scenarios. To verify this point, we compare CLHE [40] on this specific scenario with the overall product bundling scenario in Figure 3. It reveals that: In contrast to the full evaluation, ***CLHE exhibits worse performance in the Pop-to-LT setting***. This is due to its lack of explicit consideration of item popularity knowledge, which hinders its ability to achieve optimal performance in this more challenging and non-doctrinaire setting. Jointly considering these discoveries with the specific definition of this scenario, we have intuitively sparked two bold questions: Are the pre-extracted item-level user feedback features responsible for the performance decline of current bundling methods in this scenario? Is the effectiveness of various modality-aware item representations inconsistent in the Pop-to-LT scenario? To answer these questions, we conduct a surgical analysis to explore the effectiveness of each view in the Pop-to-LT scenario. From Figure 3, we can observe that: (1) CLHE w/o ui outperforms CLHE in the Pop-to-LT scenario, demonstrating the ***low quality of item-level user feedback in Pop-to-LT scenario***. Intuitively, the incorporation of the unbalanced user feedback knowledge may force the bundle constructor to predict the popular items, thereby aggravating the inherent popularity bias in bundle construction. (2) Comparing CLHE, CLHE w/o mm, and CLHE w/o bi, we notice that the ***item representations from the bundle-item view and multimodal view are indispensable for Pop-to-LT product bundling***. To sum up, all these findings collectively constitute the motivation of our work: Effectively utilizing task-related bundle-item affiliation information and popularity-irrelevant multimodal knowledge to counteract the popularity bias caused by the low-quality user-item information in existing product bundling methods.

4 Methodology

To elaborate on the technical architecture of the proposed Distilled Modality-Oriented Knowledge Transfer (DieT) framework, we first outline the product bundling backbone to illustrate the implementation process of this task. Inspired by the discoveries in Section 3, we provide in-depth definitions of the Popularity-free Collaborative Distribution Modeling (PCD) and Unbiased Bundle-aware Knowledge Transferring (UBT). Finally, we detail the optimization and discussion of DieT.

4.1 Product Bundling Backbone

Without loss of generality, the proposed DieT framework is model-agnostic and can be seamlessly integrated into any existing multimodal product bundling architecture. Inspired by the powerful hierarchical encoding capabilities on both the feature level and the item level of CLHE [40], we adopt it as the backbone of our DieT.

Item Feature Extraction. Specifically, we first extract the textual feature $\boldsymbol{t}_i$ [26] and the media feature $\boldsymbol{m}_i$ [58] for item i and then generate the fused content feature $\boldsymbol{c}_i = \text{average}(\boldsymbol{t}_i, \boldsymbol{m}_i)$. In parallel, we initialize a learnable item embedding $\boldsymbol{v}_i$ to capture item-bundle-level affiliation patterns and employ LightGCN [12] to obtain the item-level user feedback $\boldsymbol{p}_i$.

Multi-modality Fusion. Then, we concatenate these features from the same item i to obtain the final item representation $\boldsymbol{F}_i = \text{concat}(\boldsymbol{W}_c\boldsymbol{c}_i, \boldsymbol{W}_p\boldsymbol{p}_i, \boldsymbol{v}_i), \boldsymbol{F}_i \in \mathbb{R}^{3\times d}$. Here, concat$(\cdot)$ denotes the concatenation function, $\boldsymbol{W}_p$ and $\boldsymbol{W}_c$ represent the corresponding transformation matrices respectively. Subsequently, a multi-layer self-attention encoder is utilized to capture the high-order intra-item interactions among these heterogeneous modalities:

$$\begin{cases} \boldsymbol{A}_i^{(l)} = \frac{1}{\sqrt{d}}\widehat{\boldsymbol{H}}_{\boldsymbol{i}}^{(l-1)}\boldsymbol{W}_I^K(\widehat{\boldsymbol{H}}_{\boldsymbol{i}}^{(l-1)}\boldsymbol{W}_I^Q)^\top, \\ \widetilde{\boldsymbol{H}}_{\boldsymbol{i}}^{(l)} = \text{softmax}(\boldsymbol{A}_i^{(l)})\widehat{\boldsymbol{H}}_{\boldsymbol{i}}^{l-1}, \end{cases} \tag{1}$$

where l represents the l-th attention layer, $\boldsymbol{W}_I^K \in \mathbb{R}^{d\times d}$ and $\boldsymbol{W}_I^Q \in \mathbb{R}^{d\times d}$ denote the linear projections that transpose the final item representation $\boldsymbol{F}_i$ into key and value spaces, $\widehat{\boldsymbol{H}}_{\boldsymbol{i}}^{(0)} = \boldsymbol{F}_i$ and $\widetilde{\boldsymbol{H}}_{\boldsymbol{i}}^{L}$ denotes the medium feature of item i after L layers of the self-attention operator. Then we can generate the item's representation $\boldsymbol{f}_i \in \mathbb{R}^d$ by averaging the multiple features:

$$\boldsymbol{f}_i = \text{average}(\widetilde{\boldsymbol{H}}_{\boldsymbol{i}}^{(L)}). \tag{2}$$

After obtaining the item representation $\boldsymbol{f}_i$, we build another bundle-level self-attention module to capture the representation $\boldsymbol{e}_b$ of the given partial bundle b_x. Similar to Eq. 1, we use a Z-layer self-attention encoder to process the sequence of item representations $\boldsymbol{f}_i$ in the partial bundle b_x. Two trainable matrices $\boldsymbol{W}_B^K \in \mathbb{R}^{d\times d}$ and $\boldsymbol{W}_B^Q \in \mathbb{R}^{d\times d}$ project these item embeddings into the key and value spaces. After that, we can obtain the final representation of the bundle $\boldsymbol{e}_b = \text{average}(\widetilde{\boldsymbol{E}}_{\boldsymbol{b}}^{(Z)})$ by averaging the representations from each layer, where $\widetilde{\boldsymbol{E}}_{\boldsymbol{b}}^{Z}$ denotes features' representation after Z layers of the self-attention operator.

4.2 Popularity-free Collaborative Distribution Modeling

Building upon the previous empirical analyses in Section 3, we have noticed that the knowledge from different perspectives has varying effects in the Pop-to-LT scenario, and bundle-item affiliation knowledge plays a crucial role in this setting. We have also pointed out that a vast number of items exhibit diverse popularity distribution in the user-item view (infrequent) and the bundle-item view (frequent), such as niche fashion outfit generation or music collections. For these items that frequently occur within bundles, the dynamic optimization of their representation during the

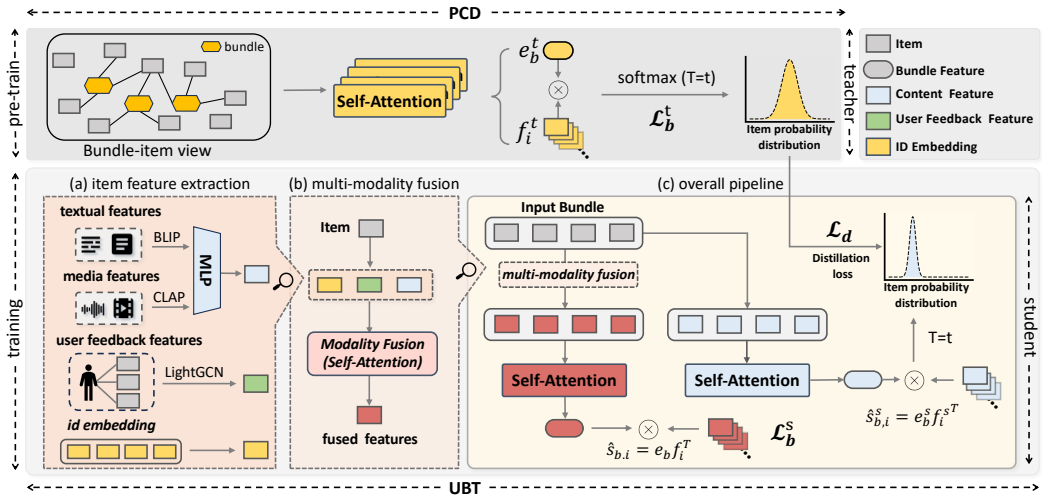


Fig. 4. The overall structure of DieT. For a given bundle, the model derives its representation from the items it contains and performs a matching process with the candidate set. Our method adopts a teacher-student distillation framework, consisting of two main modules. The PCD extracts the popularity-free bundle-item affiliation to guide the UBT for promoting long-tail items via a tailored knowledge distillation paradigm. In the student model, (a) is the feature extraction process, (b) is the multi-modal feature fusion of the items, and (c) is the overall pipeline of the student model.

training process can precisely capture the inherent bundle-item level affiliation knowledge for these long-tail items in Pop-to-LT scenarios. To capitalize on this insight, we propose the Popularity-free Collaborative Distribution Modeling (PCD) module, designed to explicitly extract and prioritize popularity-independent knowledge by modeling the collaborative distribution exclusively within the bundle-item view.

Specifically, we first initialize the teacher-specific item representation $\boldsymbol{f}_i^t$[4] of item i with the bundle-level item embedding $\boldsymbol{v}_i$. Then, we model the bundle representation $\boldsymbol{e}_b^t$ of the given partial bundle b_x via the following Eq. 3 to capture the bundle-item affiliation knowledge:

$$\begin{cases} \boldsymbol{A}_b^{(z)} = \frac{1}{\sqrt{d}} \widehat{\boldsymbol{E}}_b^{(z-1)} \boldsymbol{W}_B^K (\widehat{\boldsymbol{E}}_b^{(z-1)} \boldsymbol{W}_B^Q)^\intercal, \\ \widetilde{\boldsymbol{E}}_b^{(z)} = \text{softmax}(\boldsymbol{A}_b^{(z)}) \widehat{\boldsymbol{E}}_b^{z-1}, \end{cases} \tag{3}$$

where $\widehat{\boldsymbol{E}}_b^{(0)} = \text{concat}(\{\boldsymbol{f}_i^t\}_{i \in b})$, and z represents the z-th layer of bundle-level self-attention. The architecture comprises a total of Z attention layers, where the depth Z is a hyperparameter optimized via empirical. Subsequently, we average the multiple features to obtain the bundle representation $\boldsymbol{e}_b^t = \text{average}(\widetilde{\boldsymbol{E}}_b^{(Z)})$. After these three steps, we can compute the matching scores, which indicate the likelihood of item i being included in the partial bundle b_x to form a complete bundle via the inner product as:

$$\widehat{Logits}_{b,i}^t = \boldsymbol{e}_b^t (\boldsymbol{f}_i^t)^\intercal. \tag{4}$$

While relying solely on bundle-item level features makes it struggle to outperform existing product bundling methods (see Section 3.3), it effectively retains the bundle-item affiliation information

[4]We use superscript t to denote variables in PCD and s in UBT for differentiation.

through this simple yet universal design, thereby supporting the bundle representation modeling in transferring.

4.3 Unbiased Bundle-aware Knowledge Transferring

Intuitively, arbitrarily discarding the information of bundles from any view may lead to a defective bundle representation modeling process, thereby resulting in decreased product bundling performance. In contrast, the empirical analyses employed in Section 3.3 reveal that certain views may mislead the optimization process in the specific scenario of Pop-to-LT product bundling. This prompts us to consider efficiently utilizing the popularity-free collaborative knowledge to boost Pop-to-LT product bundling tasks without excessively impacting overall product bundling. Thus, we propose an Unbiased Bundle-aware knowledge Transferring module (UBT) to solve this issue.

Specifically, within the UBT module, we first leverage a cross-modality mapping layer to generate the student-specific content feature $\boldsymbol{c}_i^s$ for item i using the pre-extracted textual feature $\boldsymbol{t}_i$ and media features $\boldsymbol{m}_i$, which is formulated as:

$$\boldsymbol{c}_i^s = \text{average}(f_{\mathcal{T}}(\boldsymbol{t}_i), f_{\mathcal{M}}(\boldsymbol{m}_i)), \tag{5}$$

where the mapping functions $f_{\mathcal{T}}(\cdot)$ and $f_{\mathcal{M}}(\cdot)$ can provide subtle modifications for both textual and multimedia features to enable them to accommodate the product bundling task. Subsequently, we utilize the self-attention modality fusion method to generate the comprehensive item representation $\boldsymbol{f}_i^s$ with the content features $\boldsymbol{c}_i^s$, the item-level user features $\boldsymbol{p}_i$, and the bundle-item affiliation features $\boldsymbol{v}_i$. Then, we leverage the same bundle-level self-attention function in Section 4.1 to model the bundle representation $\boldsymbol{e}_b^s$ of the partial bundle b_x. These learned representations are utilized to optimize the product bundling task as the primary construction loss in Eq. 10.

Guided by the empirical findings in Section 3.3, we further refine the utilization of the content knowledge in UBT to zoom in on the Pop-to-LT product bundling task. Specifically, the content knowledge of each item is entirely popularity-free, as it only deals with multimodal information, making it entirely unrelated to the item's popularity in the user-item view. Leveraging this property, we follow the knowledge distillation paradigm to transfer the bundling-related parts of collaborative knowledge in the bundle-item view into the content features, thereby obtaining more robust and accurate content representations for Pop-to-LT scenarios. That is, we utilize content information to construct the item representation $\boldsymbol{f}_{i_c}^s = \boldsymbol{c}_i^s$ and modality-specific self-attention function to aggregate bundle representation $\boldsymbol{e}_{b_c}^s$. As in Eq. 4, we calculate the matching scores of UBT as:

$$\widehat{Logits}_{b,i}^s = \boldsymbol{e}_{b_c}^s (\boldsymbol{f}_{i_c}^s)^\top. \tag{6}$$

It is noteworthy that when calculating the scores in UBT, we exclusively employ the modality-specific bundle representation $\boldsymbol{e}_{b_c}^s$, to avoid UBT simply replicating the results of PCD, which would degrade overall performance. More importantly, the significance of popularity-free features is further enhanced, thereby effectively counteracting the bias inherent in item-level user feedback features. After these steps, the crucial question is: How do we design the knowledge transfer method to assimilate the intrinsic bundle-item structural knowledge from PCD? We convert items' matching scores in Eq. 6 into the corresponding probability distribution as follows:

$$\begin{cases} P_t = \text{softmax}(\frac{\widehat{Logits}_{b,i}^t}{T}), \\ P_s = \text{log_softmax}(\frac{\widehat{Logits}_{b,i}^s}{T}). \end{cases} \tag{7}$$

Notably, we do not enforce UBT to match the probability distribution of PCD exactly. On the contrary, we desire UBT to learn the popularity-free and bundle-familiar knowledge from PCD via

knowledge distillation. Therefore, we employ soft targets [16] to achieve this primary goal. The distillation loss can be described as:

$$\mathcal{L}_d = \frac{1}{|\mathcal{B}|}\mathrm{KL}(P_s||P_t)T^2, \tag{8}$$

where $|\mathcal{B}|$ represents the batch size, $\mathrm{KL}(\cdot)$ refers to the Kullback-Leibler (KL) divergence [15] between the probability distributions P_s and P_t of the PCD and UBT. Here, the temperature parameter T^2 is introduced in the loss computation process to normalize the gradient magnitudes of the distillation loss. We also consider a feature-based distillation loss to replace the distribution-based one, thereby transferring the feature-level bundle-item affiliation knowledge from PCD to UBT. Formally, this distilled loss can be defined as:

$$\mathcal{L}_d = \frac{1}{|\mathcal{N}|}\sum_{i=0}^{\mathcal{N}-1}\mathcal{L}(\mathrm{sim}(\boldsymbol{e}_b^t, \boldsymbol{e}_{b_c}^s)), \tag{9}$$

where $\mathrm{sim}(\cdot)$ represents the general similarity function (e.g., cosine, Euclidean, KL divergence, dot product, etc.). The primary focus of DieT is the logits distillation, while the effectiveness of feature-based distillation is verified in Section 5.3.1.

4.4 Optimization

To clarify the overall training procedure of our framework, we summarize the optimization steps in Algorithm 1. The process consists of two phases: pre-training the teacher model to capture popularity-free collaborative signals, and distilling this knowledge into the student model.

Given the item representation $\boldsymbol{f}_i$ of item i and the bundle representation $\boldsymbol{e}_b$ of partial bundle b_x, we compute the corresponding matching score $\hat{s}_{b,i} = \boldsymbol{e}_b\boldsymbol{f}_{\boldsymbol{i}}^{\mathrm{T}}$ via the inner-product function.

Following the classical bundle construction work [40], we leverage the negative log-likelihood loss to optimize the product bundling task:

$$\mathcal{L}_b = -\frac{1}{|\mathcal{B}||\mathcal{I}|}\sum_{b\in\mathcal{B}}\sum_{i\in\mathcal{I}} y_{b,i}\mathrm{log_softmax}_b(\hat{s}_{b,i}). \tag{10}$$

Then, the overall objective function $\mathcal{L}$ can be formulated as a linear combination of the construction loss $\mathcal{L}_b$, the knowledge distillation loss $\mathcal{L}_d$ and the regularization term as:

$$\mathcal{L} = \mathcal{L}_b + \lambda\mathcal{L}_d + \beta||\boldsymbol{\theta}||_2^2, \tag{11}$$

where λ denotes the weight of the knowledge distillation loss $\mathcal{L}_d$. The term $||\boldsymbol{\theta}||_2^2$ represents the $L2$ regularization term, where $||\boldsymbol{\theta}||$ denotes all the trainable parameters in our DieT.

4.5 Model Discussion

CLHE [40] is a pioneering work in the field of product bundling and achieves the SOTA performance. However, its accuracy is significantly lower when predicting long-tail items compared to popular ones (*i.e.,* the Pop-to-LT scenario), which may be attributed to the mechanical multi-view knowledge fusion. In contrast, DieT proposes a knowledge distillation framework to more effectively utilize bundle-item affiliation information and popularity-independent content information to mitigate the low-quality user-item information in existing bundles, thereby reducing popularity bias. Notably, DieT achieves significant performance improvements in the Pop-to-LT scenario while also delivering remarkable gains in the overall bundling scenario with the meticulous design of these two modules.

While DieT is empirically driven, its effectiveness in mitigating popularity bias can be theoretically interpreted through the lens of signal denoising and regularization constraints. We posit that the supervision signal from the User-Item view ($\mathcal{D}_{ui}$) comprises two entangled components: intrinsic user preference (S_{int}) and popularity noise (S_{pop}). Standard models trained on $\mathcal{D}_{ui}$ tend to overfit

Algorithm 1 Training Procedure of DieT Framework

Require: User-item interaction matrix $\mathbf{D}$, Item textual/media features $\mathcal{T}, \mathcal{M}$, Bundle sets B.
Require: Hyper-parameters: Temperature T, distillation weight λ, learning rate η.
Ensure: Optimized parameters of the Student model (UBT) θ_S.

// **Phase 1: Pre-training the Teacher Model (PCD)**
Initialize Teacher parameters θ_T.
while not converged **do**
  for each batch of bundles $b \in B$ **do**
    Derive item representations f_i^t using bundle-item ID embeddings (Eq. 2).
    Compute bundle representation e_b^t via self-attention (Eq. 3).
    Calculate prediction logits $\hat{y}_{b,i}^t = e_b^t (f_i^t)^\top$ (Eq. 4).
    Compute construction loss $\mathcal{L}_b^t$.
    Update $\theta_T \leftarrow \theta_T - \eta \nabla \mathcal{L}_b^t$.
  end for
end while
Freeze Teacher parameters θ_T.
// **Phase 2: Training the Student Model (UBT)**
Initialize Student parameters θ_S.
while not converged **do**
  for each batch of bundles $b \in B$ **do**
    // **Teacher Forward (Guidance)**
    Compute teacher logits $\hat{y}_{b,i}^t$ using frozen θ_T.
    Compute teacher distribution $P_t = \text{softmax}(\hat{y}_{b,i}^t / T)$ (Eq. 7).
    // **Student Forward**
    Fuse multimodal features c_i^s, user feedback p_i, and ID embeddings v_i to get f_i^s.
    Compute student logits $\hat{y}_{b,i}^s$ and distribution $P_s = \text{softmax}(\hat{y}_{b,i}^s / T)$.
    // **Loss Computation**
    Calculate construction loss $\mathcal{L}_b$ with ground truth (Eq. 10).
    Calculate distillation loss $\mathcal{L}_d = T^2 \cdot \text{KL}(P_s || P_t)$ (Eq. 8).
    Total loss $\mathcal{L} = \mathcal{L}_b + \lambda \mathcal{L}_d + \text{Reg}$ (Eq. 11).
    // **Optimization**
    Update $\theta_S \leftarrow \theta_S - \eta \nabla \mathcal{L}$.
  end for
end while
return θ_S

S_{pop} due to its dominance in the data distribution, leading to popularity bias. In contrast, the Bundle-Item view ($\mathcal{D}_{bi}$), constructed based on item compatibility (often by experts or semantics), is largely independent of user-side interaction volume. Consequently, the Teacher model (PCD), trained exclusively on $\mathcal{D}_{bi}$, captures a distribution P_T driven primarily by intrinsic compatibility (S_{int}), effectively serving as a popularity-free prior.The knowledge distillation objective, $\mathcal{L}_{KD} = KL(P_S || P_T)$, acts as a regularization constraint on the Student model (UBT). By forcing the Student's output distribution P_S to align with P_T, we penalize the Student for learning high-probability patterns solely derived from S_{pop} (which are absent in P_T). Mathematically, this minimizes the mutual information between the Student's predictions and the popularity bias signal, effectively transferring the structural invariance of the unbiased view to the final model.

Table 1. Statistics of two real-world bundling datasets. "#" denotes the count or average of corresponding entities: users (#U), bundles (#B), items (#I), and their interactions (e.g., #B-I: bundle-item interactions). "#Avg." represents average values (e.g., #Avg.I/B indicates the average number of items per bundle).

Dataset	#U	#I	#B	#B-I	#U-I	#Avg.I/B	#Avg.B/I	#Avg.I/U	#Avg.U/I
Spotify	118,899	72,453	12,309	359,935	21,032,433	29.24	4.97	176.89	290.29
POG	2,311,431	31,217	29,686	105,775	6,345,137	3.56	3.39	2.75	203.26

5 Experiments

To evaluate our proposed approach, we conduct extensive experiments on two real-world datasets. In particular, we aim to answer the following research questions:

- **RQ1:** How does DieT perform against the SOTA methods on the full product bundling task and the Pop-to-LT bundling scenario?
- **RQ2:** Is DieT still effective and robust with diverse distillation strategies, popularity ratio, and bundling scenarios?
- **RQ3:** Does DieT demonstrate the generalization with other base product bundling backbones?
- **RQ4:** What is the computational complexity and resource efficiency of DieT compared to other baseline methods?
- **RQ5:** How does DieT improve Pop-to-LT bundle construction in practice, and what qualitative evidence supports its advantages over baseline models?

5.1 Experimental Settings

We first introduce the two datasets used in this work, followed by a discussion of general product bundling methods and how traditional debiasing approaches in recommendation are adapted for bundling tasks. Finally, we provide a detailed explanation of the implementation of this study.

5.1.1 Datasets. The application scenarios for product bundling are diverse, including e-commerce, meal, and gaming platforms, *etc.*[4, 6]. To evaluate our method, we select two datasets from the fashion domain POG and the music playlist dataset Spotify because they encompass the three types of features mentioned earlier. Additionally, to evaluate the effectiveness of our method in Pop-to-LT scenarios, we utilize user-item interaction data to measure popularity. Items are ranked according to their interaction frequency, and a certain proportion of high-popularity items from the head and low-popularity items from the tail are selected. Specifically, while keeping the bundle composition unchanged, we only modify whether items act as observed inputs or prediction targets in the test set, which enables a controlled evaluation under popularity shift without altering bundle semantics. For the Spotify dataset, where playlists are typically long, we further sample a fixed-length subset of items from each playlist to construct partial-bundle prediction instances. In particular, each sampled bundle contains 20 items, among which 10 items are used as inputs and the remaining 10 items are treated as prediction targets. This procedure is applied consistently in both the standard and Pop-to-LT evaluation settings to ensure sufficient coverage and stable comparison. For each dataset, we randomly divide all bundles into training, test, and validation sets in a 7:2:1 ratio. The details of the datasets are presented in Table 1.

5.1.2 Baselines. Considering that product bundling has been less explored, we observe a scarcity of research that precisely matches our framework. Therefore, we select several leading methods and tailor them to our specific settings. For a fair comparison, all baseline methods utilize three types of features as our method and employ the identical negative log-likelihood loss.

Additionally, we incorporate debiasing methods from traditional recommendation tasks into the best baseline and compare them with our method.

General Product Bundling Methods:

- **MultiDAE** [59] adopts an auto-encoder architecture, where each item is encoded into a fixed-size vector. To generate a bundle representation, it utilizes average pooling to aggregate the individual item representations.
- **MultiVAE** [29] is a variational auto-encoder model built upon MultiDAE, incorporating variational inference techniques to probabilistically model item representations, allowing for more flexible and robust bundle representations.
- **Bi-LSTM** [11] models each bundle as a sequence of items and employs bidirectional LSTM to capture contextual relationships between items in both forward and backward directions, generating a comprehensive bundle representation.
- **Hypergraph** [63] represents each bundle as a hypergraph, with items as nodes and the bundle as a hyperedge. A GCN is used to aggregate features, capturing higher-order item relationships for the bundle representation.
- **Transformer** [57] customizes a transformer architecture to effectively capture item interactions within a bundle. By leveraging self-attention mechanisms, it models complex relationships among items and generates an expressive bundle representation.
- **CLHE** [40] integrates three types of data and employs a two-level attention mechanism to effectively capture hierarchical information for learning bundle representations. By introducing contrastive learning enhancement, CLHE effectively handles modality missing and noise issues, thereby improving model robustness and the quality of bundle representations.

Debiased Product Bundling Methods:

- **IID** [67] encourages item representations to remain neutral to popularity across all directions by minimizing the similarity between each direction and the popularity vector during training. In our task, we minimize the aggregated similarity between all directions and the popularity vector to mitigate popularity bias. Meanwhile, we introduce orthogonal regularization, which enforces orthogonality among item representation directions to ensure that each direction encodes only independent feature information.
- **PID** [60] suggests removing the popularity component from item representations by isolating a popularity-related direction during training, and neutralizing this vector during inference. In the bundling task, we fix the last column of item representations as the popularity vector during training and reconstruct the popularity vector through other directions to isolate it. During inference, we set the popularity vector to zero.

5.1.3 Implement Details. Following the pioneering work [40], we define the embedding size and the dimension of hidden representations as 64. We use Xavier initialization [8], Adam optimizer [22], and batch size 2048 for DieT and the compared baselines. We apply L2 regularization using the weight_decay parameter of the Adam optimizer, which corresponds to standard weight decay. The regularization is applied to all learnable parameters, including item embeddings. Although weight decay affects the embedding vectors of items not present in a given batch, we intentionally retain this behavior, as it helps avoid the uncontrolled growth of rarely updated embeddings and improves generalization in sparse settings. We embark on a grid search to identify the optimal hyperparameter settings. Specifically, we explore a range of learning rates within $\{10^{-2}, 10^{-3}, 5\times10^{-3}, 10^{-4}, 5\times10^{-4}\}$, and adjust β within the range of $\{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$. In most cases, optimal performance is achieved when β is set to 10^{-5}. In POG, the optimal learning rate is 10^{-3}, while in Spotify, it is 10^{-4}. As for the various hyper-parameters used in contrastive learning and data augmentation, we adhere to the optimal settings established in prior works. In knowledge distillation, we explore

Table 2. Performance comparison between DieT and baselines on Pop-to-LT product bundling scenario on two datasets. "%Improv." denotes the relative improvement over the strongest baseline. The best baselines are underlined. The p-value represents the result of a significance test. A p-value less than 0.05 indicates a significant result, while a p-value less than 0.01 indicates a very significant result.

Dataset	Spotify				POG			
Metric	R@20	N@20	R@40	N@40	R@20	N@20	R@40	N@40
MultiDAE	0.0172	0.0125	0.0339	0.0199	0.0377	0.0160	0.0508	0.0189
MultiVAE	0.0121	0.0086	0.0252	0.0144	0.0570	0.0285	0.0755	0.0326
Bi-LSTM	0.0385	0.0320	0.0621	0.0426	0.0121	0.0055	0.0228	0.0078
Hypergraph	0.0614	0.0539	0.1000	0.0713	0.0551	0.0238	0.0725	0.0277
Transformer	0.0166	0.0119	0.0319	0.0187	0.0460	0.0262	0.0636	0.0301
CLHE	0.0741	0.0645	0.1168	0.0837	0.0661	0.0392	0.0904	0.0446
IID	0.0737	0.0649	0.1136	0.0828	0.0707	0.0417	0.0911	0.0470
PID	0.0745	0.0649	0.1159	0.0835	0.0691	0.0419	0.0925	0.0467
DieT	**0.0860**	**0.0772**	**0.1283**	**0.0959**	**0.0773**	**0.0430**	**0.0989**	**0.0479**
%Improv.	**15.43**	**18.95**	**9.85**	**4.58**	**9.34**	**2.62**	**6.91**	**1.91**
p-value	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.05	< 0.05

the temperature T within the range of $\{1, 2, 3\}$ and the parameter λ for distillation loss within the range of $\{0.5, 1.0, 1.5, 2.0\}$. In most cases, optimal performance is achieved with $T = 2$ and $\lambda = 1.0$. During the pre-training of the teacher model, we set $T = 1$ to allow it to complete the bundling task normally. During the training of the student model, we set $T = 2$ to enable the teacher model to better guide the training of the student model.

We follow the designs specified in their respective papers for the baseline models to achieve optimal performance. As for general product bundling methods other than CLHE, we also use the three modalities mentioned above and combine them into item representations using an MLP. Regarding Debiased Product Bundling methods, as highlighted in the related work, there is a lack of research on addressing the popularity bias of items within bundles. In this study, we explore several traditional debiasing methods from recommendation systems and attempt to apply them to the product bundling task.

5.2 Performance Comparison (RQ1)

We first compare DieT with several baselines on popular ranking metrics under the Pop-to-LT and full product bundling scenarios using two real-world datasets. Next, we introduce a popularity evaluation metric to provide a more intuitive demonstration of our method's effectiveness in addressing popularity bias.

5.2.1 Performance on Pop-to-LT scenario and Overall scenario. To demonstrate DieT's effectiveness, we conduct tests from both the Pop-to-LT and full product bundling scenario perspectives, comparing them with multiple baselines on real-world datasets from two domains. As illustrated in Table 2 and Table 3, we highlight the best results on the same datasets with bold font and underline the best baselines. Popular ranking metrics such as Recall@K and NDCG@K are employed [5]. Our observations are as follows:

[5]For simplification, we leverage N@k and R@k to indicate the NDCG@k and Recall@k in the following figures and tables.

Table 3. Performance comparison between DieT and baselines on full product bundling scenario on two datasets. "%Improv." denotes the relative improvement over the strongest baseline. The best baselines are underlined. The p-value represents the result of a significance test. A p-value less than 0.05 indicates a significant result, while a p-value less than 0.01 indicates a very significant result.

Dataset	Spotify				POG			
Metric	R@20	N@20	R@40	N@40	R@20	N@20	R@40	N@40
MultiDAE	0.0614	0.0541	0.0971	0.0700	0.3082	0.2008	0.3367	0.2074
MultiVAE	0.0534	0.0470	0.0836	0.0605	0.3076	0.1891	0.3346	0.1954
Bi-LSTM	0.0411	0.0348	0.0675	0.0467	0.3141	0.1972	0.3423	0.2039
Hypergraph	0.0733	0.0619	0.1149	0.0806	0.2695	0.1815	0.2888	0.1861
Transformer	0.0672	0.0590	0.1078	0.0772	0.3035	0.1846	0.3244	0.1895
CLHE	0.0815	0.0694	0.1286	0.0905	0.3811	0.2776	0.4090	0.2842
IID	0.0812	0.0697	0.1290	0.0911	0.3820	0.2704	0.4092	0.2769
PID	0.0796	0.0673	0.1253	0.0878	0.3815	0.2754	0.4086	0.2818
DieT	**0.0832**	**0.0711**	**0.1322**	**0.0930**	**0.3842**	**0.2779**	**0.4113**	**0.2843**
%Improv.	**2.09**	**2.01**	**2.48**	**2.09**	**0.11**	**0.11**	**0.51**	**0.04**
p-value	< 0.01	< 0.05	< 0.01	< 0.05	-	-	< 0.05	-

1) **In the Pop-to-LT scenario, which is the main focus of our work, DieT outperforms all the baselines on all datasets.** As illustrated in Table 2, the performance of DieT outperforms several product bundling methods and some debiasing methods in traditional recommendation, demonstrating the effectiveness of our popularity-free collaborative distribution modeling and unbiased bundle-aware knowledge transferring.
2) **Improvement of DieT to the backbone correlates with the average number of items per bundle.** This indicates that PCD can extract more comprehensive bundle-item relationship information in datasets with denser bundle-item affiliation, making it more effective in guiding UBT. Hence, DieT shows notable improvement in both scenarios on Spotify. The improvement of DieT at k=20 is greater than that at k=40, indicating our method can more accurately identify the most relevant long-tail items in smaller recommendation lists.
3) **DieT achieves comparable results to the optimal baseline model overall**. It is attributed to UBT's well-designed knowledge transfer mechanism, which leverages content features to integrate popularity-free collaborative knowledge from PCD, retaining UBT's strong overall performance while achieving some improvements in the Pop-to-LT scenario.
4) **The performance of DieT in the Pop-to-LT scenario surpasses that of several conventional debiasing techniques applied to CLHE.** We evaluate several viable debiasing methods from traditional recommendation systems, adapting them for our product bundling task. The findings indicate that although these methods partially alleviate popularity bias in the bundling construction process, they are not entirely suited for product bundling and are outperformed by DieT. Furthermore, the IID method incurs substantial computational costs, as it processes features for each individual item, resulting in prohibitive runtime increases on large-scale datasets.
5) **The results of the experiment are significant.** We further conduct a t-test between DieT and CLHE, the experimental results show that in the Pop-to-LT setting, the relative improvement is mostly very significant (p-value < 0.01). Although the p-values for R@40 and N@40 in the POG dataset are slightly greater than 0.01, still less than 0.05, we believe that the overall performance

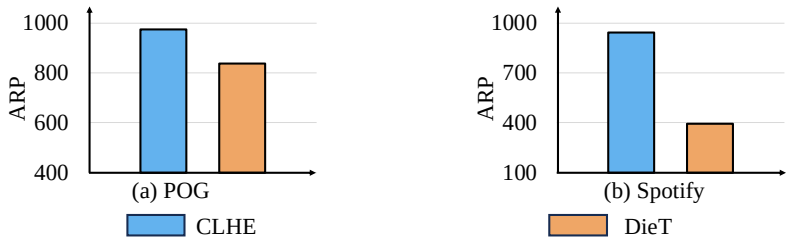


Fig. 5. The results of DieT and CLHE in the Pop-to-LT scenario, where the vertical axis (ARP) represents the average popularity of recommended items in each bundle, measured by the number of times an item appears in the training data. This metric evaluates the effectiveness of mitigating popularity bias across two datasets. The blue bars represent CLHE results, while the orange bars correspond to DieT results. Subfigures (a) and (b) illustrate the outcomes on the Spotify and POG datasets, respectively.

in the Pop-to-LT scenario is statistically significant. Furthermore, in the overall scenario with the Spotify dataset, we are confident that most improvements are substantial (p-value < 0.05) and achieve comparable performance on the POG dataset. This experimental discovery is consistent with our initial description that the primary contribution of our paper lies in the significant improvements in the Pop-to-LT setting, with comparable results in the overall bundling scenario.

5.2.2 Analysis of Popularity Debiasing. Following [45, 68], we evaluate the accuracy of our DieT and other baselines within Pop-to-LT scenarios to reflect the popularity bias, the results are shown in Table 2. Given our tailored preprocessing of the dataset in the Pop-to-LT scenario, the construction targets are exclusively long-tail items. The improved performance of our approach in this context highlights its exceptional capability in tackling long-tail challenges, effectively completing bundles with long-tail items, while maintaining result accuracy. Furthermore, to further demonstrate the ability of our method in addressing the popularity bias issue, we introduce the Average Recommendation Popularity (ARP) [10] as a supplementary evaluation metric, which calculates the average popularity of predicted items in each bundle as the popularity bias metric. Experimental results in Figure 5 show that DieT predicts items with a lower ARP, indicating a lower average popularity of items in the completed bundle which is aligned with the motivation of our DieT. Notably, the decline in ARP is more pronounced on the Spotify dataset, aligning with the conclusion drawn in Section 5.2, analysis 2). DieT demonstrates superior performance in datasets with a higher number of bundle-item interactions.

5.3 In-depth Analyses of DieT (RQ2)

In the first part, we analyze the overall performance of DieT. Additionally, to further evaluate its effectiveness and robustness under diverse distillation strategies, varying popularity ratios, and different bundling scenarios, we conduct targeted experiments and perform in-depth analyses of these three aspects.

5.3.1 Effectiveness Analysis of Diverse Distillation Methods. As we introduce in Section 4.4, there are multiple distillation methods. Here, we employ different distillation methods within our DieT framework to evaluate the effectiveness of each approach. The results in Table 4 demonstrate that regardless of the distillation method employed, UBT equipped with knowledge significantly improves in the Pop-to-LT scenario. Different methods achieve varying improvements. However, simply stacking these methods does not yield better results. The reason lies in their different optimization objectives and constraints, which lead to inconsistencies when used simultaneously,

Table 4. Results of diverse knowledge distillation methods in DieT on the dataset of POG. Here, FD stands for feature distillation, and LD stands for logits distillation.

Method	R@20	N@20	R@40	N@40
CLHE	0.0661	0.0392	0.0904	0.0446
DieT (LD)	**0.0773**	0.0430	**0.0989**	0.0479
DieT (FD)	0.0705	**0.0459**	0.0950	**0.0513**
DieT (FD+LD)	0.0730	0.0416	0.0893	0.0452

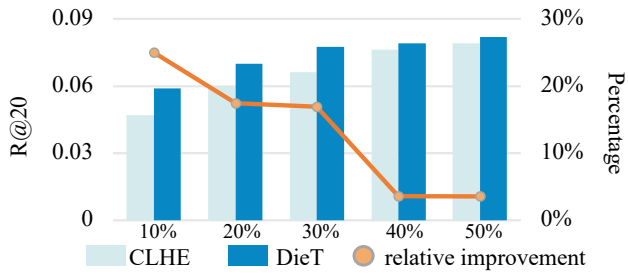


Fig. 6. Performance comparison between DieT and SOTA method CLHE on diverse popularity ratios on POG. The horizontal axis corresponds to the selection of popular items at various ratios. The bar chart displays the recall@20 scores for both models, and the line chart shows the percentage increase of DieT compared to CLHE at each ratio.

thus affecting the overall effectiveness. Therefore, to save computational resources, we opted to use Logits distillation. Here, we simply elucidate that our approach proves effective under various knowledge distillation methodologies. As for more detailed investigations, such as various loss calculations, additional optimization techniques, or how to better integrate the advantages of both distillation methods, we leave those for future work.

5.3.2 Robustness Analysis on Diverse Popularity Ratios. In Table 2, taking into account both the impact of popularity and the size of the test set, we categorize items based on their popularity rank into the top-30% popular and the top-30% long-tail items. Items in the middle 40% range are excluded to ensure a clear separation between the two groups and avoid ambiguity in popularity attribution. For all other popularity thresholds used in our experiments, we follow the same rule and discard bundles containing middle-range items during testing. As a result, only a subset of bundles satisfies the constraints for each scenario. It should be noted that as the definition of popularity becomes stringent, the number of bundles meeting the criteria decreases. Under this setting, we select five different popularity intervals in the test set and compare the performance of the backbone with our model. As the popularity requirements increase, Figure 6 clearly shows a noticeable performance drop in the backbone model, highlighting the challenge in more demanding scenarios, while the DieT model experiences a smaller decline and shows a certain improvement relative to the backbone model. At the same time, we represent the improvement percentage with orange lines across different popularity proportions. As the selection criteria become stricter, the improvement of our model becomes more apparent. This further underscores the effective knowledge transfer between the two modules in DieT, which helps mitigate popularity bias.

5.3.3 Robustness Analysis on Diverse Bundling Scenarios. Here, we intend to delve deeper into the performance of DieT across the various scenarios we previously analyzed. As shown in Table

Table 5. Performance comparison between DieT and SOTA method CLHE on diverse bundling scenarios on POG dataset. The best results are highlighted in bold.

Scenario	overall		LT-to-Pop		Pop-to-Pop		Pop-to-LT	
Model	CLHE	DieT	CLHE	DieT	CLHE	DieT	CLHE	DieT
R@20	0.3811	**0.3842**	**0.8571**	0.8566	0.4247	**0.4312**	0.0661	**0.0773**
N@20	0.2776	**0.2779**	0.6326	**0.6334**	0.2949	**0.2962**	0.0392	**0.0430**
R@40	0.4090	**0.4113**	0.8763	**0.8832**	0.4688	**0.4693**	0.0904	**0.0989**
N@40	0.2842	**0.2843**	0.6373	**0.6500**	0.3046	**0.3049**	0.0446	**0.0479**

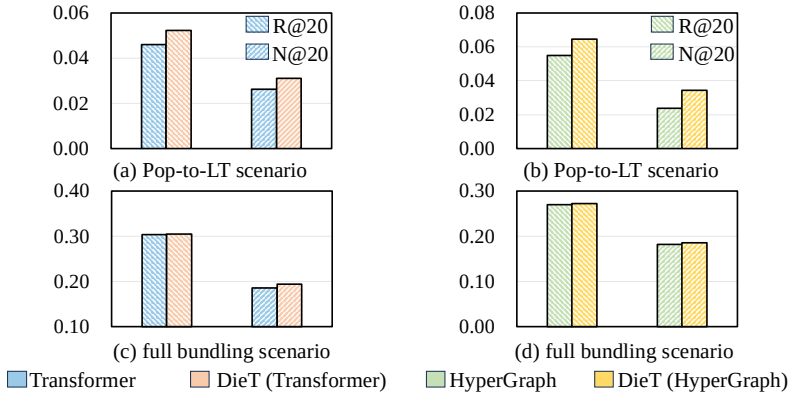


Fig. 7. Generalization analysis of applying DieT to other backbones of Transformer and Hypergraph. Different colors represent different models, as shown in the legend. The horizontal axis corresponds to various metrics, while the vertical axis indicates their respective values.

5, we compare the results of DieT and CLHE across these four different scenarios. Firstly, it is evident that the model's performance naturally surpasses random overall results when predicting high-popularity items. This is particularly evident in the LT-to-Pop scenario. The results indicate that DieT has achieved significant improvements in the Pop-to-LT scenario while obtaining results comparable to CLHE in other scenarios. This proves that our carefully designed knowledge transfer strategy maintains the strength of the UBT and integrates the PCD's capability to handle long-tail items, making it a competitive solution across various application scenarios.

5.4 Generalization Analysis of DieT (RQ3)

To further demonstrate our approach's effectiveness, we modify several baseline methods that show promising results. We integrate our framework into the Transformer and HyperGraph methods. We keep the other parts of the baseline unchanged. Consistent with the previous description, we use a model with a self-attentive bundle-item affiliation excavator as the PCD, and these two models with three types of features as UBK's backbone. We compare the results of the models before and after incorporating DieT in terms of Pop-to-LT and full bundling scenario performance. The results in Figure 7 reveal that even in relatively simple models, our approach achieves significant improvements in the Pop-to-LT scenario and some enhancements in full bundling scenario performance. This indicates that DieT is a modality-oriented knowledge transfer framework independent of specific models, demonstrating the Generalization of our approach across different models.

Table 6. Computational complexity comparison between DieT and baseline on Spotify and POG. "par." denotes parameters, "tra." and "inf." denote training time and inference time for one epoch.

Datasets	model	#par. (M)	GPU(MB)	#tra. (s)	#inf.(s)
Spotify	CLHE	62.35	1,114	15.56	6.85
	DieT (CLHE)	106.17	1,734	19.06	6.65
	Transformer	62.44	850	13.17	6.15
	DieT (Transformer)	99.73	942	13.77	6.22
	HyperGraph	62.46	1,085	14.87	8.13
	DieT (HyperGraph)	99.53	1,413	16.60	10.26
POG	CLHE	28.99	1,270	20.61	8.67
	DieT (CLHE)	57.98	1,695	24.85	8.45
	Transformer	29.07	1,090	16.39	10.52
	DieT (Transformer)	45.25	1,124	17.01	9.74
	HyperGraph	29.08	1,090	15.20	7.10
	DieT (HyperGraph)	45.05	1,123	15.39	7.54

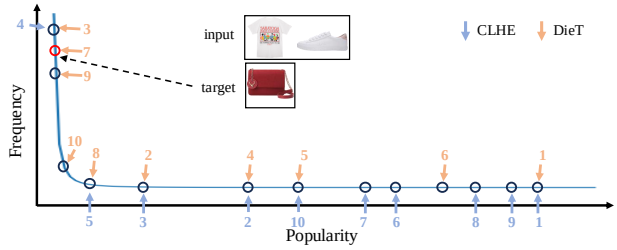


Fig. 8. The popularity distribution of top-10 items predicted by DieT and the baseline CLHE for several given popular items, with numbers representing the score ranking of the items. The blue curve represents the simulated item popularity distribution. The x-axis indicates popularity, creasing along the axis direction, while the y-axis shows the frequency of item occurrence.

5.5 Computational Complexity Analysis of DieT (RQ4)

To demonstrate the lightweight characteristics of our proposed DieT, we compare the computational complexity between DieT and the backbones (e.g., CLHE, Transformer, and HyperGraph) on POG and Spotify on NVIDIA Tesla V100. As shown in Table 6, we observe that due to the characteristics of the knowledge distillation paradigm, the additional computation of the teacher model inevitably increases the memory requirements and training time. It is worth noting that DieT does not conduct the knowledge distillation in the inference stage. Therefore, its online inference complexity is comparable to CLHE in magnitude, making it highly efficient in practical applications. Additionally, we conduct a complexity analysis of Transformer and HyperGraph, comparing them with the application of the DieT framework. The findings align with the conclusions drawn from the comparison between CLHE and DieT, which further demonstrates the lightweight characteristic, and robustness of our DieT.

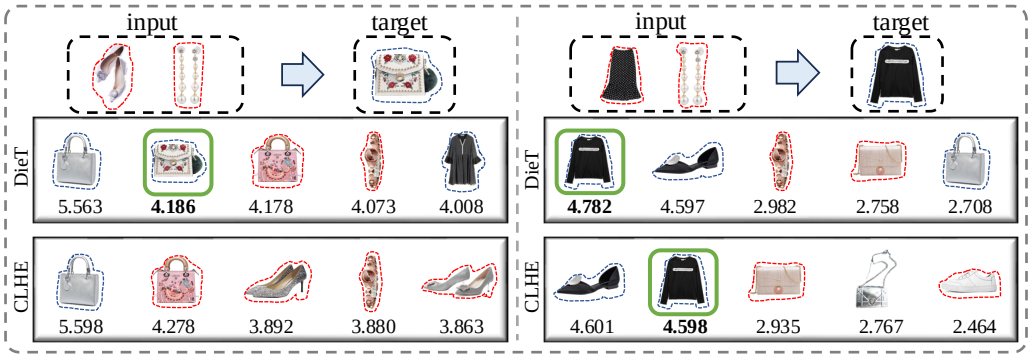


Fig. 9. The top-5 items and their scores predicted by DieT and CLHE for several given popular items, with red and blue dashed lines marking the popular and long-tail items. The digits displayed below each item indicate the model's scores for those items.

5.6 Case Study (RQ5)

In the previous sections, we carry out several experiments and provide detailed analyses of the quantitative results. Here, we further present case studies to analyze how DieT enhances Pop-to-LT bundle construction and provides qualitative insights compared to the baseline models. Specifically, we select the top 30% items by popularity as popular items and the bottom 30% as long-tail items, and examine DieT's performance from two complementary perspectives.

5.6.1 Case Study on Popularity Distribution of Results. In Figure 8, we use a distribution curve to approximate the item popularity distribution. As the x-axis increases, item popularity grows, while the y-axis represents the frequency of items with that popularity. This indicates that highly popular items are relatively fewer compared to less popular ones in the dataset. As shown in the legend, the arrows in two colors represent the positions of the top-10 items predicted by each model on the popularity distribution curve. Compared with CLHE, DieT successfully identifies the target item within the top-10 predictions in this case. It is evident that CLHE tends to favor highly popular items for bundle construction, as indicated by the blue arrows being concentrated in the high-popularity region. In contrast, DieT distributes its predictions more evenly across the popularity spectrum, demonstrating a fairer selection of items and a better balance between popular and long-tail products.

5.6.2 Case Study on Comparative Analysis of Prediction Scores. From another perspective, Figure 9 presents two cases to demonstrate the scores of the top-5 items ranked by CLHE and DieT in the Pop-to-LT scenario. We also use red dashed lines to denote popular items and blue dashed lines to represent long-tail items. Compared to CLHE, DieT achieves higher scores for low-popularity target items. In the left example, although neither model predicts the target long-tail item as the top-1, DieT successfully predicts the target item within the top-5 items. In the right example, DieT outperforms CLHE with higher prediction scores for the target item, placing it higher in rank. Additionally, as marked, DieT's constructed bundles include more long-tail items (indicated by a greater number of items within the blue dashed lines), demonstrating its superior ability to promote long-tail items in the Pop-to-LT scenario.

6 Conclusion and Future Work

In this work, we systematically investigated bundle construction issues to achieve combinations better aligned with real-world applications by addressing inherent popularity biases. By pairing popular items with niche or newly introduced ones, we enhance the visibility and promotion of long-tail products. We analyzed existing issues and challenges, highlighting uncertainties from user feedback data and suboptimal performance in Pop-to-LT scenarios. To address these, we proposed the Distilled Modality-Oriented Knowledge Transfer framework (DieT), which outperforms leading techniques on datasets from two domains. By leveraging a pre-trained teacher model to guide the training of the student model, DieT facilitates better collaboration between popularity-free bundle-item features and multimodal content features. This approach mitigates the inherent popularity bias in user behavior and effectively promotes long-tail items. Extensive experiments on two domain-specific datasets reveal that DieT surpasses leading approaches in the Pop-to-LT scenario and achieves competitive performance overall. We also carried out various experiments and detailed analyses, which validate the efficacy of our proposed knowledge distillation framework. Furthermore, while DieT is primarily tailored for Pop-to-LT bundle completion, its core paradigm exhibits broader generalizability. Within the bundling domain, it serves as a viable solution for extreme Long-Tail to Long-Tail (LT-to-LT) cold-start scenarios, where historical user interactions are entirely absent. Beyond explicit bundling, this debiasing mechanism can be seamlessly adapted to general complementary item retrieval tasks (e.g., retrieving visually or functionally compatible items in e-commerce search scenarios). This ensures that cross-item recommendations are driven by intrinsic item compatibility rather than merely historical click-through popularity.

Despite its effectiveness, DieT has practical limitations that guide our future research. The multi-layer attention and distillation mechanisms make scaling to massive item spaces computationally expensive. To enable practical deployment and address the lack of personalization, future systems can adopt a decoupled architecture where DieT serves as an offline engine to pre-compute debiased representations, while a lightweight online ranking module handles efficient vector retrieval and injects user context. Advancing this personalized ranking, however, is currently hindered by a severe lack of publicly available multimodal datasets with explicit user histories. We plan to construct such datasets to support models that dynamically balance popular and niche items based on individual user tolerances, explicitly capturing complex relationships via graph methods. Beyond personalization, we aim to optimize user feedback extraction in an end-to-end manner rather than relying on pre-extracted features. Since successful bundling often spans multiple domains like books and music, future models must also better capture semantic cross-domain connections. Finally, developing adaptive hyperparameter tuning strategies remains an important step to improve the framework's overall stability.

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